

Written exam 16 December, 2022

Course title: Network Optimization

Course no: 42115

Aids allowed: All

Exam duration: 4 hours

Weighting: The points are indicated at each question

Language: You can write in Danish or English

Figures: Figures can be found on the last pages of this assignment. You can use them in your answer.

The Pisinger cake

The Pisinger cake was created by Oscar Pischinger, a famous baker from Vienna, in the 19th century.



The preparation consists of the following eight tasks:

- A. Leave the wafer-sheets to cool under load.
- B. Coat the wafer-sheets with the obtained cream.
- C. Using a round bowl, cut the wafers to get 10 round sheets.
- D. Peel and fry almonds.
- E. Cream: Beat eggs, chocolate, egg yolks, sugar in steam, when the mixture starts to thicken, cool it a bit, then mix in the frothy butter, lemon juice and ground almonds.
- F. Chop almonds and mix them into the cream.
- G. Pour the chocolate icing over the cooled cake and keep it cool.
- H. Icing: Heat chocolate. Add oil and milk.

The duration of each task is given in the following table together with predecessors that need to be finished before the task can start

task	time (min)	predecessors
A	20	B
B	6	C, F
C	5	
D	12	C, E
E	15	
F	3	E, D
G	4	A, H
H	7	E, F, B

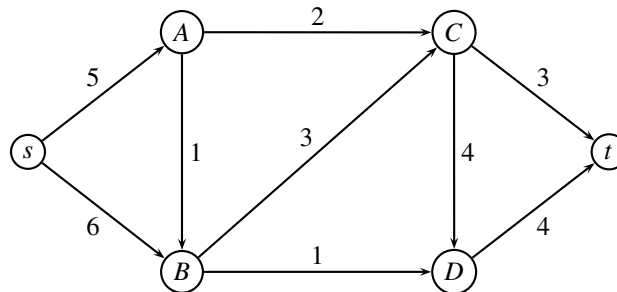
Problem 1 (1 point) Sort the tasks in topological order (either as graph, or as a list). ■

Problem 2 (3 point) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan T . Show the recursion you are using. ■

Problem 3 (1 point) Indicate the critical path, and argue why the edges are on the critical path. ■

Maximum flow problem

In the following network the edge weights denote the capacity. The problem is to find the maximum flow from s to t .



Problem 4 (4 point) Use the *preflow-push* (also called *push-relabel*) algorithm to solve the above problem. Give sufficient details to make it possible to follow the algorithm. ■

Problem 5 (1 point) Describe how you find a minimum cut using the labels from previous question, and report the cut, and its capacity. ■

Max flow min cut

We are considering a maximum flow problem defined on a directed graph $G = (V, E)$ and edge capacities $u_{ij} \geq 0$. The solution vector is denoted (x_{ij}) , and a cut is denoted $(R, \bar{R})$.

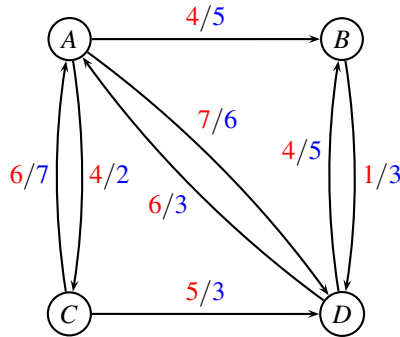
Problem 6 (4 point) Prove or disprove each statement. To disprove a statement it is sufficient to show an example.

- A) If all edge capacities in G are increased by an additive constant k , the min cut $(R, \bar{R})$ remains unchanged.
- B) If all edge capacities in G are distinct, the min cut $(R, \bar{R})$ is unique.

■

Repositioning of empty containers

Consider the following network, where costs are red, and capacities are blue.



Due to the imbalance in trade, empty containers tend to accumulate at various ports. For the above network, one could have the following surplus of containers:

node i	A	B	C	D
surplus σ_i	8	-3	4	-9

We would like to reposition the containers in the cheapest possible way. It is assumed that all empty containers are equal.

Problem 7 (1 point) Formulate the problem of repositioning containers as a *minimum cost flow problem*. ■

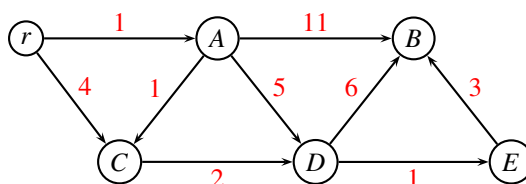
Problem 8 (2 point) Find an initial solution by solving a maximum flow problem. When solving the maximum flow problem always use the augmenting path with *largest capacity* in the residual graph. ■

Problem 9 (4 point) Use the *augmenting cycle* algorithm or *Network Simplex* to solve the problem. Report the optimal solution. ■

Shortest Path Problem in a dynamic network

In network communication there are several protocols to ensure that packages are sent to the correct destination. One of these, the *Open Shortest Path First* (OSPF), makes use of Dijkstra's algorithm to route packages in a dynamic network, where connections are added/removed.

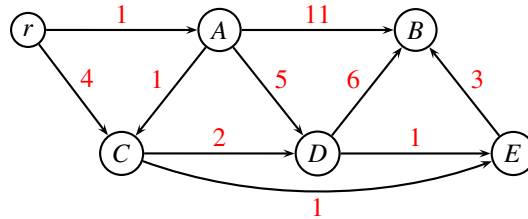
Consider the following directed graph where distances w_{ij} are marked with red.



Assume that you should find the shortest path from r to all other nodes.

Problem 10 (2 point) Solve the shortest path problem using Dijkstra's algorithm. Report the paths. ■

Now, assume that an edge (C, E) with distance 1 is added to the graph:



Problem 11 (2 point) How can we modify Dijkstra's algorithm to find the shortest path to all nodes, but somehow re-using calculations from previous questions? (You may assume that all calculations from previous questions are available). ■

Problem 12 (1 point) Use the modified algorithm to find the shortest path from r to all nodes in the new graph where edge (C, E) was added. Report the paths. ■

Transportation problem

Two producers A_1 and A_2 are delivering products to two customers B_1 and B_2 . The producers can produce at most 8 and 7 items respectively. Customer B_1 has a demand of 5 items, while customer B_2 has a demand of 4 items. The transportation costs are given by the following table

	B_1	B_2	max production
A_1	4	5	8
A_2	1	6	7
demand	5	4	

Note, that the problem is not balanced.

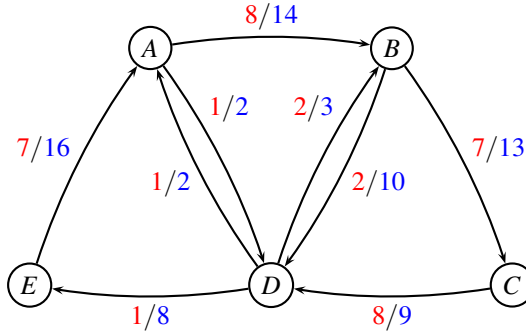
Problem 13 (1 point) Formulate the problem as a transportation problem, by specifying the network and all costs / demands. ■

Problem 14 (2 point) Find a greedy solution for the transformed problem: Consider the nodes in the order A_1, A_2 always choosing the cheapest option for transporting the production. ■

Problem 15 (4 point) Find an optimal solution using the network simplex algorithm for the transportation problem. In each iteration, give sufficient details to follow the algorithm. Describe the termination criteria, and report the optimal solution. ■

Flowing containers

MEDITLINE (Route net of a Mediterranean liner shipping company)



For edge (i, j) the first number is the **cost** c_{ij} and the second number is the **capacity** u_{ij} (in TEU).

We consider again the route net MEDITLINE from project assignment 2022. The network is *identical* to the one used in the project assignment, but we have *new demands* as follows:

k	s_k	t_k	d_k	w_k
1	D	C	3	2
2	A	D	2	4

Commodity $k = 1$ is shipping three 40 foot containers (size $w_k = 2$ TEU) from D to C , and commodity $k = 2$ is shipping two 40 foot high-cube containers (size $w_k = 4$ TEU) from A to D .

We are considering a *weighted multi-commodity flow problem*: For each commodity, a container of size w_k TEU has to follow the same path, but the d_k different containers can follow individual paths.

Problem 16 (2 point) For each commodity k in MEDITLINE, write up a table with all possible routes from s_k to t_k through the graph $G = (V, E, c, u)$. A route may only use edges where the capacity is at least w_k TEU, and it needs to be simple (i.e. cycle-free). For each commodity k and each feasible route state the corresponding cost of sending the w_k TEU. (Hint: There should be 5 distinct routes.) ■

We solve the LP-relaxed problem through *column generation*. After a few iterations we have the *restricted master problem* on the form

$$\begin{aligned}
 \min \quad & c_1 x_1 + c_2 x_2 + c_3 x_3 \\
 \text{s.t.} \quad & 2x_2 + 4x_3 \leq 14 & (y_{AB}) \\
 & & \leq 2 & (y_{AD}) \\
 & 2x_1 + 2x_2 & \leq 13 & (y_{BC}) \\
 & & 4x_3 \leq 10 & (y_{BD}) \\
 & & & \leq 9 & (y_{CD}) \\
 & & & \leq 2 & (y_{DA}) \\
 & 2x_1 & \leq 3 & (y_{DB}) \\
 & & 2x_2 & \leq 8 & (y_{DE}) \\
 & & 2x_2 & \leq 16 & (y_{EA}) \\
 & x_1 + x_2 & \geq 3 & (\bar{y}_1) \\
 & & x_3 & \geq 2 & (\bar{y}_2) \\
 & x_1, x_2, x_3 \geq 0
 \end{aligned} \tag{1}$$

Let y_{ij} denote the dual variable corresponding to the capacity constraint for edge (i, j) , and let $\bar{y}_k$ denote the dual variable corresponding to the demand constraint for commodity k .

The dual variables for the above model are: $y_{DB} = -14$, while all other are $y_{ij} = 0$. Furthermore, $\bar{y}_1 = 46$, $\bar{y}_2 = 40$.

Problem 17 (4 point) For a given commodity k describe how a route with most negative reduced cost can be found using a shortest path algorithm. Draw the network for each commodity k (nodes, edges, costs, start-node, end-node), and report the optimal solution to the shortest-path problem, as well as the reduced cost. ■

Problem 18 (1 point) Which path should be added to the restricted master problem? ■

Chocolate factories

Two chocolate factories A and B are producing at most 500 chocolate-bars per month, each. Customer C demands 350 chocolate-bars, while customer D demands 400 chocolate-bars per month. Production costs are \$5 at factory A , while production costs are \$6 at factory B .

The factories can either deliver directly to the customers, or they can deliver through a warehouse W . Shipping **costs** per chocolate-bar are as follows:

From/to	W	C	D
A	1	5	6
B	2	4	5
W	-	3	2

The **capacity** of the shipping company is given by the following table:

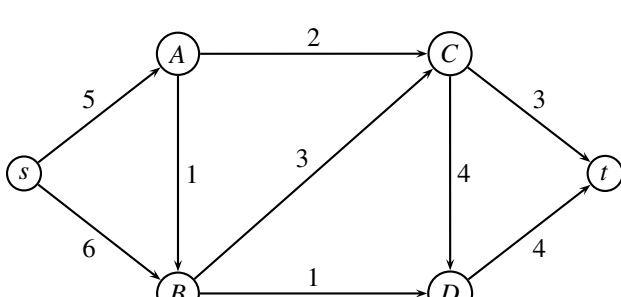
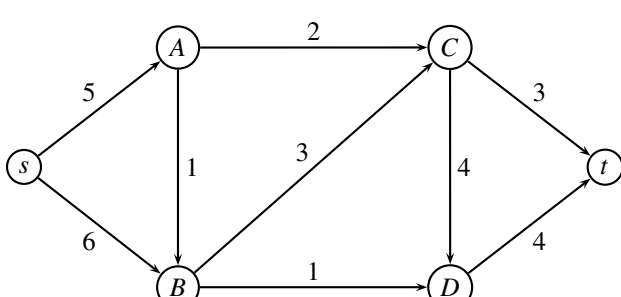
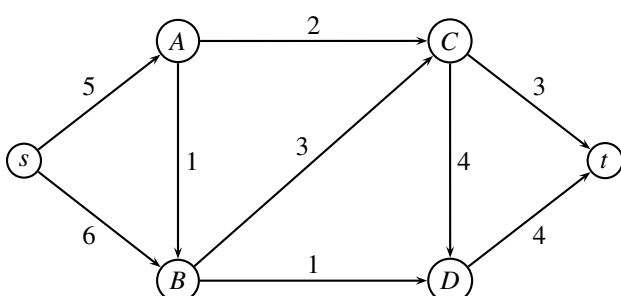
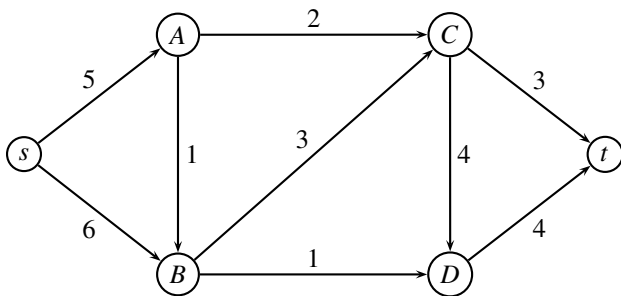
From/to	W	C	D
A	400	500	350
B	200	600	350
W	-	200	200

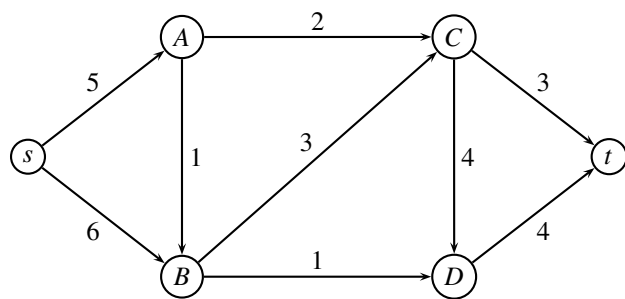
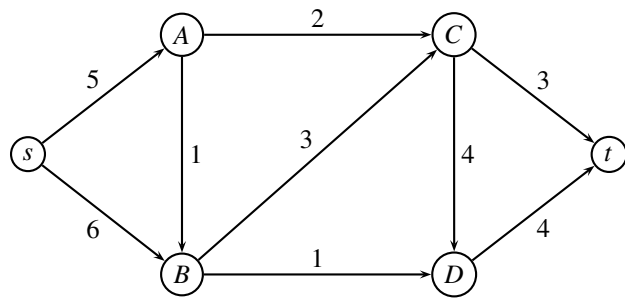
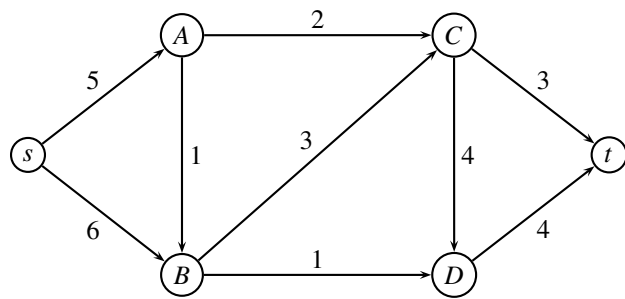
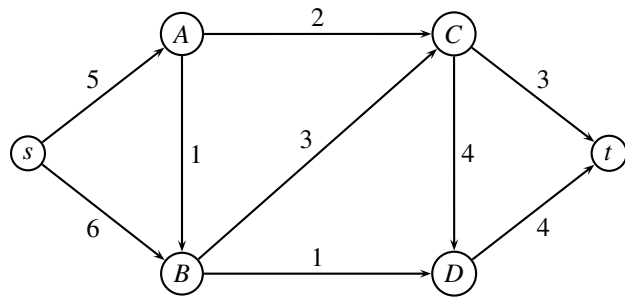
The warehouse has a limited **capacity**, so at most 300 chocolate-bars can be shipped through the warehouse. However, **at least** 80 chocolate-bars need to be shipped through the warehouse.

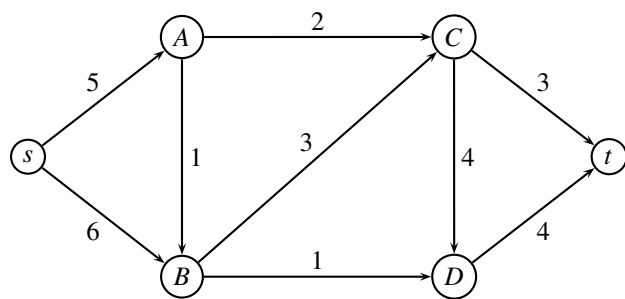
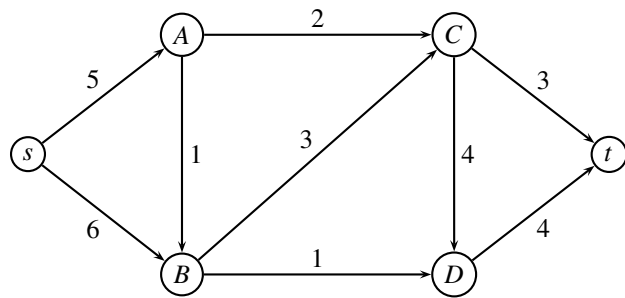
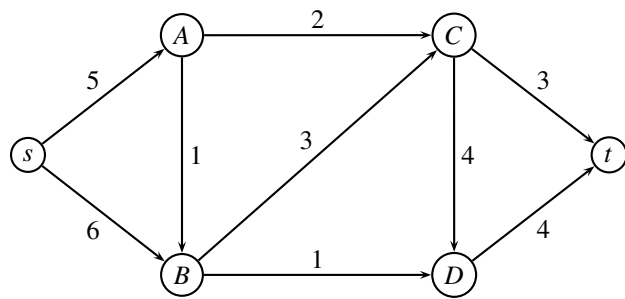
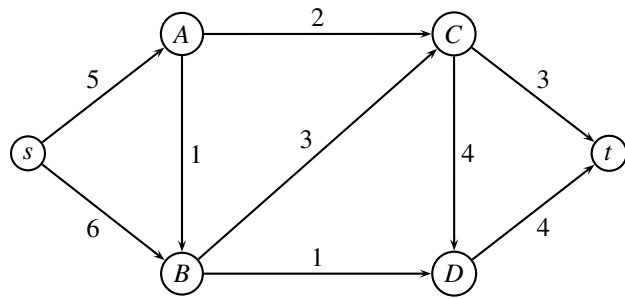
Problem 19 (5 point) Formulate the problem of deciding the company's production to minimize the total cost as a *minimum cost flow problem*. You should indicate the (**cost**, **capacity**) on *all* edges. Only upper bounds on the capacity are allowed. ■

_____ end of assignment _____

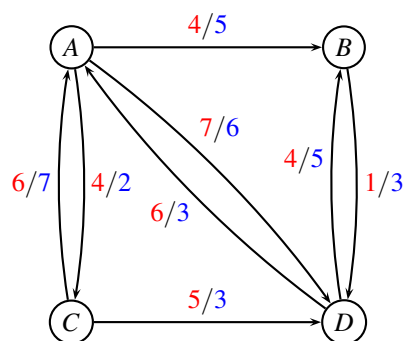
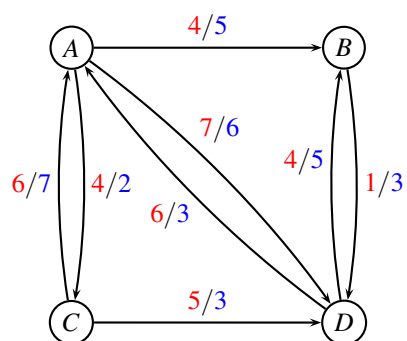
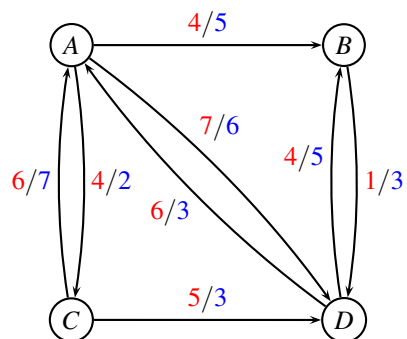
Maximum flow problem, Figures

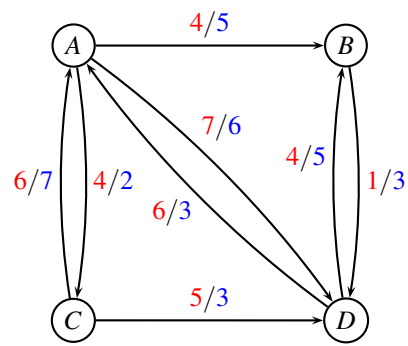
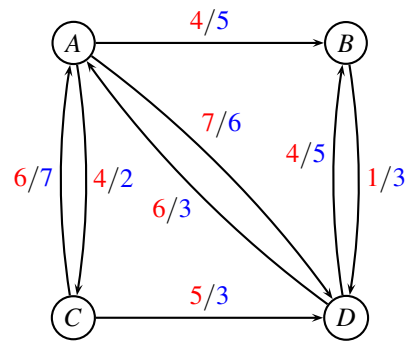
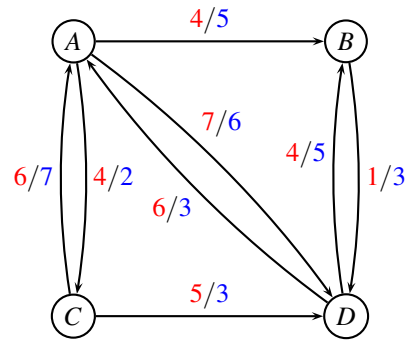






Repositioning of empty containers, Figures





Flowing containers, Figures

